

point or simply the IoT server that controls the communication between publisher and subscriber. It handles the data that the publisher transmits for other gadgets known as subscribers, as shown in Fig. 1.

For instance, if data is to be transferred from a temperature sensor to a farmer's handheld phone, the data signals are sent by the temperature sensor (publisher) while the farmer's device is the receiver (subscriber) of data.

A number of MQTT brokers are available in the free and open source space including Mosquitto, which implements the MQTT protocol for IoT based communications. It's lightweight, high-performance, and suitable for use on all devices, from low-power single boards like Arduino and ESP8266 to computers and servers. Instead of a local PC, Mosquitto can be deployed on a cloud based server that implements this broker so that IoT communication is control-

lable over the Internet.

MQTT broker applications can be installed on specialised devices such as Raspberry Pi, Arduino, laptops, servers, and many others for real world applications.

Cloud platforms for MQTT

A number of MQTT cloud broker services are available, in which the MQTT brokers are available as infrastructure on demand without any physical deployment of these.

CloudMQTT

URL: <https://www.cloudmqtt.com>

CloudMQTT is a commonly used MQTT broker and can be used for IoT interfacing that gadget communication. IoT applications using CloudMQTT include multiple scenarios like smart toll plazas, smart cities, smart homes, or smart offices.

For researchers, the CloudMQTT includes the free package named 'Cute Cat' in which five users/connections can be created and the IoT devices can be connected, as shown in Fig. 2. The free instances can be built on the CloudMQTT network,

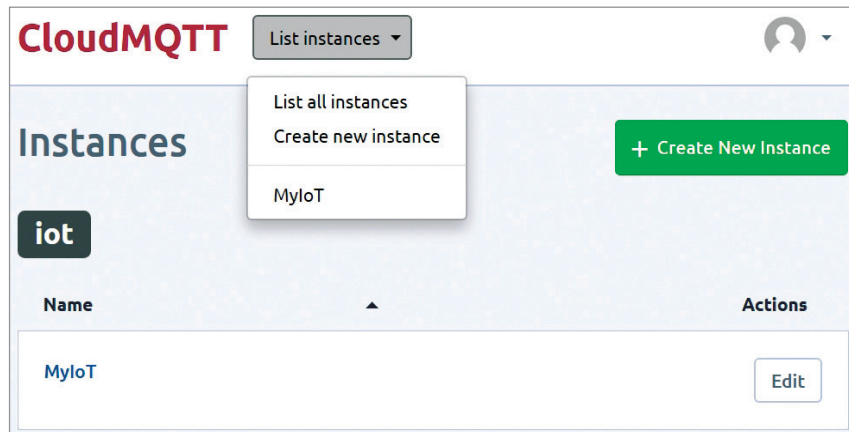


Fig. 2: Creating a free instance on CloudMQTT

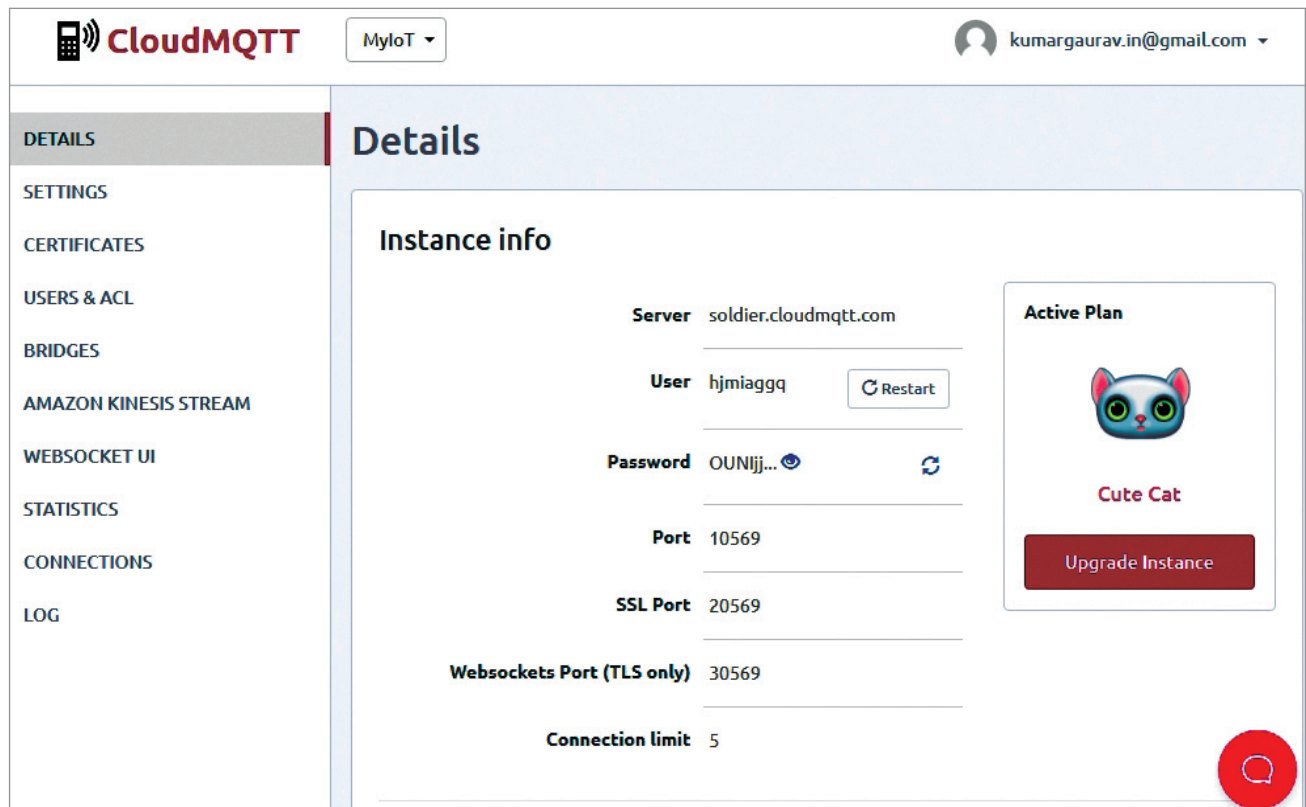


Fig. 3: Viewing authentication in CloudMQTT for connections